AI CLUSTERS

Fast switching OCS for AI and Hyperscale Data Centers: Benefits, Technologies and Emerging Applications

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Note :
OCS White Paper presentation (Optical Circuit Switching for AI and Hyperscale Datacenters: Benefits, Technologies, Current and Future Deployments) is available via OCP Online Training Academy

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OCS Proven at Scale, Faster Switching Expands its Use Cases

- **OCS is proven at scale:** Google demonstrated 30% cost and 40% power improvement by eliminating O-E-O conversion
- **Switching speed is a bottleneck:** 3D MEMS dominates OCSs but reconfigures on millisecond timescales, limiting its use to slowly changing workloads.
- **This has constrained the ecosystem:** Sub-millisecond switching has not been broadly available, so fast-lock transceivers, control planes, and orchestration software have not matured around it
- **Faster switching expands the opportunity:** Sub-millisecond OCS unlocks new all-optical use cases that require dynamic traffic steering. CPO compounds this need—more optics, closer to the servers, faster churn

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Faster OCS Drives Both Network Benefits and System-Level Challenges

Benefits	Challenges
Higher network utilization – bandwidth can be rapidly reassigned to changing traffic demands, reducing stranded capacity and improving overall fabric efficiency	**Increased operational complexity** – Dynamic topologies require more sophisticated control software, scheduling, routing, telemetry, monitoring, and troubleshooting.
Dynamic topology adaptation – network connectivity can be optimized in real time to match changing application and traffic demands.	**Advanced transceiver requirements** – Fast switching requires transceivers to rapidly acquire clock and data, restore equalization, and stabilize links after every reconfiguration.
Reduced network overbuild and oversubscription – faster reconfiguration can lower the amount of excess capacity required to accommodate traffic variability, reducing infrastructure costs	**Higher switch endurance requirements** – If fast switching enables more frequent topology changes, switching hardware must withstand significantly more switching cycles over its operational lifetime.
Enables AI-native fabrics – supports dynamic GPU interconnects, disaggregated inference, MoE communication, and workload-aware network optimization	
Reduced reliance on electrical packet switching – more traffic can remain in the optical domain, lowering packet-processing bottlenecks, latency, and power consumption	

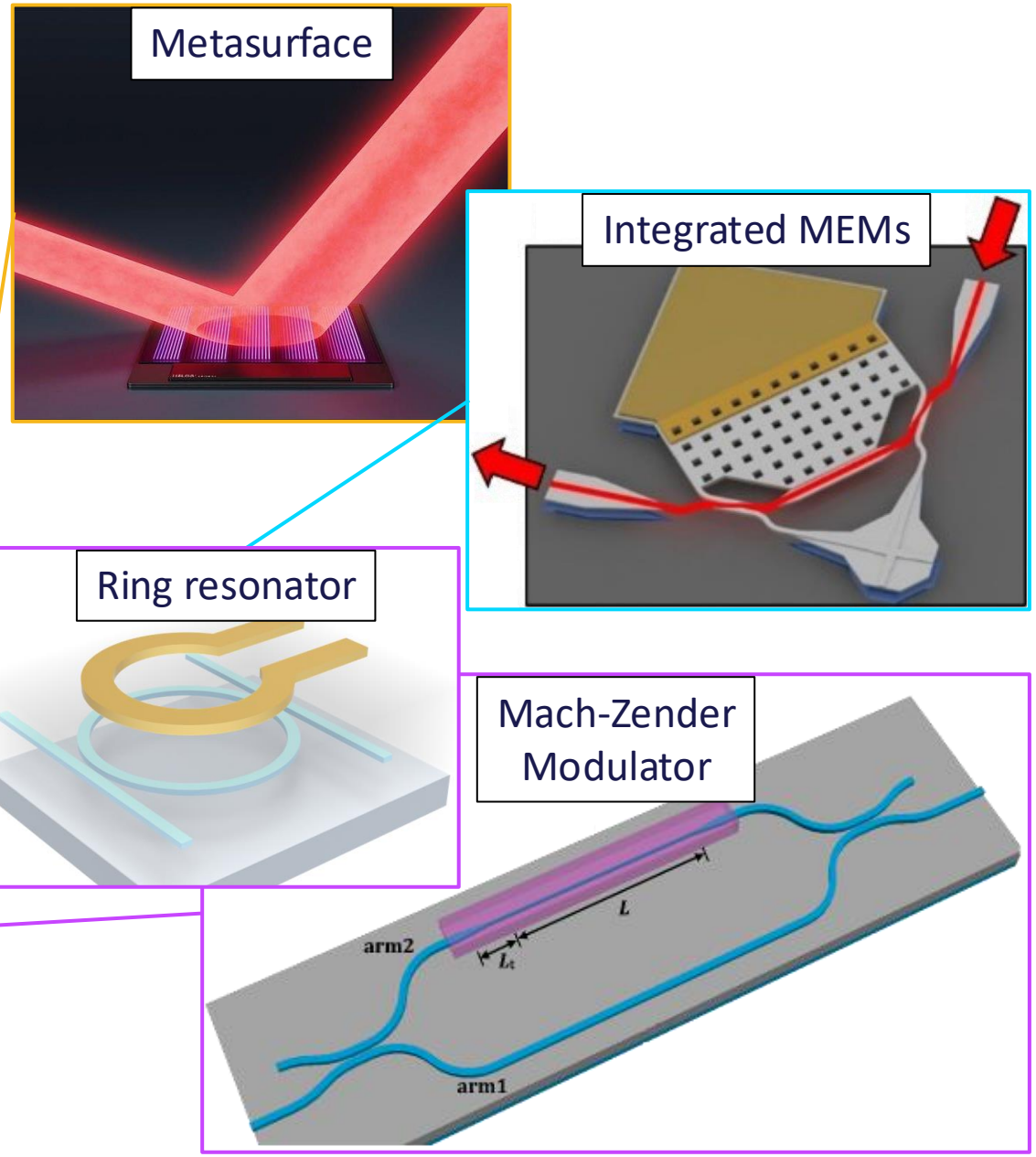
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The Leading Fast Switching Technologies are Integrated Photonics and Metasurfaces

- **Radix, cycle endurance, and insertion loss are critical system metrics outside this comparison**

Technology	Reconfiguration Time
Metasurface (free-space)	~50 µs
Integrated photonics MEMS	1–20 µs
Integrated photonics thermo-optic	10–100 µs
Integrated photonics electro-optic	ns–µs (tens of GHz)
	ns (30–56 GHz)
	sub-ns (100+ GHz – THz)

Silicon photonics



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Fast Switching OCS: Use Cases

Enable practical improvements in AI network utilization, efficiency, and scalability.

Fast-switching OCS evolves from dynamic traffic engineering to workload-aware connectivity and more flexible optical fabrics.

1 TRAFFIC ENGINEERING

Dynamic capacity allocation

PROBLEM

Traffic hotspots between rack pairs

OUTCOME

Higher network utilization

2 AI WORKLOAD ADAPTATION

Workload-aware topology

PROBLEM

Communication patterns change across training phases

OUTCOME

Better GPU efficiency

3 FAST OPTICAL FABRICS

Fine-grained scheduling

PROBLEM

Coordinating rapid optical path changes

OUTCOME

Latency + power gains

4 SCALABLE OPTICAL FABRICS

Optical Scheduling

PROBLEM

Scaling connectivity across large AI clusters

OUTCOME

Flexible, scalable AI infrastructure

Switching speed matters when it delivers better utilization, GPU efficiency, and scalable AI connectivity.

Use Case : Data-Center Traffic Engineering

Dynamically allocate optical capacity to match changing data-center traffic demand

1

PROBLEM

Capacity is stranded

Bursty, uneven traffic creates hot rack pairs while capacity elsewhere remains underutilized.

2

SOLUTION

Steer light toward demand

Fast-switching OCS dynamically reallocates optical circuits based on rack-to-rack traffic demand.

3

OUTCOME

Improve network utilization

Offload high-bandwidth flows onto optical circuits, reducing packet-fabric congestion and improving utilization.

DYNAMIC OPTICAL CAPACITY ALLOCATION

DETECT

Identify high-demand rack pairs

RECONFIGURE

Allocate optical circuits where needed

ADAPT

Adjust connectivity as traffic evolves

Fast-switching OCS enables demand-driven optical connectivity, improving utilization and adaptability.

Use Case : AI Workload Adaptation

Adapt optical connectivity to changing AI workload demands.

1

PROBLEM

One topology cannot fit every phase

AI workloads generate changing communication patterns across training and inference.

2

SOLUTION

Make topology workload-aware

Fast-switching OCS adapts optical connectivity to workload demands.

3

OUTCOME

Keep accelerators productive

More efficient communication can improve GPU utilization.

WORKLOAD-AWARE CONNECTIVITY

OBSERVE

Identify communication patterns

RECONFIGURE

Adapt optical topology

OPTIMIZE

Improve accelerator efficiency

Fast-switching OCS enables workload-aware connectivity for more efficient AI infrastructure.

Use Case : Advanced Fast Optical Fabrics

Exploring optical connectivity as a flexible, fine-grained schedulable infrastructure resource.

1

PROBLEM

Fixed topologies constrain scale

Growing AI clusters demand greater bandwidth and connectivity flexibility.

2

OPPORTUNITY

Schedule optical connectivity

Fine-grained scheduling could share optical paths as communication demands change.

3

POTENTIAL OUTCOME

Enable composable AI fabrics

Adaptable optical connectivity could support more flexible, scalable AI infrastructure.

ADVANCED FABRIC CAPABILITIES

SCHEDULE

Allocate paths on demand

COMPOSE

Connect accelerators flexibly

SCALE

Support larger AI fabrics

Fast-switching OCS opens new possibilities for schedulable, scalable AI connectivity.

Summary and Call to Action

OCS is proven at scale with large power and cost benefits, but current reconfiguration speed is limiting which AI use cases it addresses. This problem requires coordination across the ecosystem.

Call to action to realize fast-switching OCS:

- Define workload-level benchmarks
- Standardize end-to-end reconfiguration metrics
- Align transceivers, control planes, and OCS requirements

Learn more and get involved:

- OCS Whitepaper: https://www.opencompute.org/documents/ocp-ocs-white-paper-april-2026-final-pdf
- Fast Switching Whitepaper: Coming soon
- OCS Working Group: https://ocp-all.groups.io/g/ocp-optical-circuit-switching/topics

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Thank You!



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